Bruno Moya Ruiz

AI-NATIVE SOFTWARE ENGINEER | AI AGENTS, PYTHON, TYPESCRIPT, MCP

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PROFESSIONAL SUMMARY

AI-native software engineer and technical founder with 5+ years building production software end to end: interface, backend, data and infrastructure. Python/FastAPI and TypeScript/React on PostgreSQL, specialised in **agentic AI systems**: LLM orchestration, tool and function calling, **MCP integrations** and a clear boundary between model reasoning and deterministic execution. As CTO of Topbrokers I lead engineering on a production platform where I built from scratch the **remote farm of 140+ Android phones**, run workflows with daily peaks above **1,000 conversations** and cut infrastructure costs from **~EUR 2,500 to ~EUR 600 a month**. A decade at Glassear: AR hardware, computer vision and Glassear Studio, a platform where official coding agents build, test and fix XR apps. I build every day with coding agents, delegating execution but not architecture. Machine-readable profile: brunomoya.com/llms.txt

CORE COMPETENCIES

AI agents & agentic workflows | LLM orchestration, tool calling & MCP | Python & FastAPI backends | TypeScript, React & Node.js | Distributed systems: RabbitMQ, Celery, PostgreSQL | LLM observability: Langfuse, OpenTelemetry, Sentry | Computer vision & XR (Unity, OpenXR, WebXR) | SaaS products & Stripe payments

WORK EXPERIENCE

Topbrokers

Jul 2025 - Present

CTO | AI Systems & Backend Engineering · Mataró, Spain - Remote

Joined Jul 2025; CTO since Nov 2025. Team of three engineers (up to four), reporting to the CEO.

- **Built from scratch the remote device farm**: a distributed system to remotely control **140+ Android phones**, enabling native WhatsApp operations, operational monitoring and messaging automation at scale. Designed and delivered end to end; the platform's only greenfield subsystem.
- **Lead engineering on an established production AI automation platform** for real estate, evolving an inherited system across AI agents, LLM workflows, WhatsApp automation, REST APIs, CRM integrations, distributed scraping, remote Android control and cloud infrastructure.
- **Develop Python/FastAPI services on PostgreSQL** and **event-driven workflows on RabbitMQ and Celery** coordinating AI agents, customer data, devices, messaging workloads and internal tools.
- Develop LLM and WhatsApp workflows processing thousands of inbound conversations, with daily peaks above **1,000 conversations**; develop and operate **whatsapp-srv**, the production messaging service in **TypeScript/Node.js**.
- Develop the **internal CRM in Flutter**, the cross-platform client real-estate agents use every day against the platform APIs.
- Wired **Sentry** into the core WhatsApp service and **connected Codex to Sentry over MCP**, so an agent reads recent production errors, traces them to the code and proposes fixes for review.
- Built a **visual journey editor** (React Flow, FastAPI, PostgreSQL) with a chat sandbox that runs WhatsApp AI flows step by step against the OpenAI API before release, plus a journey viewer to monitor live runs and retry failed ones.
- Built an **MCP server for the in-house task platform** (Python, FastMCP, Supabase RPCs; 20 tools) so Claude Code and Codex can read, create, comment on and close tasks; packaged a recurring multi-repo change as a **Codex plugin skill**.
- **Reduced cloud infrastructure costs from ~EUR 2,500/month to ~EUR 600/month** through architecture redesign, server optimisation and migration, while improving monitoring, logging, health checks and alerts.
- Production systems support customers including **Comprarcasa, Housfy and AIC**, serving hundreds of real-estate professionals.

Glassear

May 2015 - Present

Founder & CEO | Software & AI Engineer · Barcelona, Spain

Selected among ACCIÓ's 50 highest-potential Catalan startups (2025).

- Built **Glassear Studio** (studio.glassear.com), a **fully agentic platform** for building XR apps from natural language: the user describes the app, and an official coding agent (Claude Code or OpenAI Codex) builds it, tests it and fixes it.
- Moved Studio in 2026 from an in-house AI Planner and Intent Graph to an **orchestrator of official coding agents**: the provider's agent owns planning and the tool loop; Studio owns sessions, authentication, isolation, observability and validation.
- Built a common **provider adapter layer** over official SDKs, ACP and structured CLIs: Codex and Claude Code in production, plus GitHub Copilot, Cursor, OpenCode, Kimi, Mistral Vibe and Kiro behind feature flags.
- Gave agents **100+ MCP tools** to create, build, lay out and test apps, including simulated clicks, gaze, grabs and frame capture, so an agent **checks its own work in a private XR preview**; ship a **remote MCP server** (Streamable HTTP with OAuth) for ChatGPT, Claude and other MCP clients.
- Built Studio end to end as a **TypeScript monorepo**: React IDE with Monaco, **Fastify API on PostgreSQL** with WebSocket streaming, a WebXR simulator on three.js/IWSDK and a bridge that drives the Unity runtime through structured scene patches; **200+ test files** enforce cross-layer contracts.
- Instrumented agent runs with **Langfuse over OpenTelemetry**: a trace per run, token usage (cached vs. uncached, normalised across Codex and Claude) and cost, and failed runs flagged as errors.
- Agents run in a **Docker sandbox** with an egress proxy and per-run ephemeral MCP tokens, or locally through **Glassear Agent**, built in **Rust** (workspace crates for protocol, security, transport, storage, processes and providers) and shipped in a **Tauri 2** desktop app.
- Built **Glassear One** (patent pending), a smartphone-powered optical see-through AR headset, across multiple prototype generations; its first use case is **GlassMusic**.
- Built the company site in Next.js, later rebuilt in Astro; run a self-hosted **n8n** instance for production automation of glassear.com and glassmusic.app.

Freelance (Ciane Consulting | evalua-lo.com)

Sep 2024 - Feb 2025

Full-Stack Product Engineer · Remote

- **Built Hess Test**, a psychological assessment **SaaS** for HR teams, end to end: product concept, public website, paid product and delivery, with **React** on the frontend and **PHP/WordPress** on an **MVC** backend.

- **Implemented Stripe** as the payment layer and **integrated the OpenAI API** to turn each completed questionnaire into a plain-language summary alongside the study-based scoring.
- Designed database, user flows, automated emails, scoring logic and processing queues within **shared-hosting constraints**.

Patient73 / FECAMM
Jul 2022 - Jan 2025

Technical Lead · Barcelona, Spain

- Managed the technical direction and accessibility requirements of a therapeutic games platform for patients with rare diseases (technical management and coordination, not hands-on implementation).
- Organised two international Game Jams: the first remotely from Montreal (**10+ games, EUR 3,500+ in prizes**); the second month-long with universities, RCD Espanyol and hospitals (**100+ games, EUR 11,000 in prizes**, ceremony at Barcelona City Hall).

ManoMotion
Dec 2018 - Jun 2019

AR / Computer Vision Developer Intern · Stockholm, Sweden

- Developed AR applications with the ManoMotion SDK and improved gesture interaction with OpenCV and neural-network-based tracking, while still in secondary education.

Computer Vision Center
May 2018 - Jul 2018

Computer Vision / Android Developer Intern · Barcelona, Spain

- Developed a native Android app in Java for real-time pose, face and object tracking, integrating OpenCV and neural networks on-device.

SELECTED PROJECTS

GlassMusic **In development**
AR sheet music for musicians on Glassear One: head-tilt measure scrubbing, palm-anchored score, a teacher/student web platform and optical measure detection as a Celery job. Supported by the UAB-Cruïlla Chair of Music and Technology; shown at 4YFN / MWC 2025.
Unity, ARFoundation, MediaPipe, FastAPI, Supabase, Celery, React

JobAgent
Agentic job-search system: role discovery, semantic matching, CV tailoring and browser automation on a hybrid LLM + deterministic architecture with MCP, structured outputs and human-in-the-loop approval.
Python, FastAPI, Celery, PostgreSQL, Playwright, React, TypeScript

TalentGapAnalyzer **Winner, UAB The Hack 2025**
AI workforce analytics with OpenAI, Claude and Gemini fallback, caching, token/cost tracking, bias detection and explainable outputs.
Python, FastAPI, Pandas, scikit-learn, React

NotionAI-MyMind **284 stars | 700+ users**
Open-source second brain on Notion with automatic image tagging by computer vision (InceptionV3 / Clarifai) in 2020; browser extension and Flutter app on Google Play.
Python (Quart), TensorFlow, Flutter, Docker

MapAnim / Route Motion Studio
Deterministic map animation engine: preview and export share one scene function; geodesic math, cinematic camera and headless rendering to H.264, ProRes 422 and alpha overlays.
TypeScript, React, MapLibre GL, Supabase, Playwright, FFmpeg, Vitest

SeiVoice **HackUPC 2024**
Voice ordering agent for supermarkets: two LangChain ReAct agents with custom tools, fuzzy product matching and persistent memory.
Python, FastAPI, LangChain, Flutter, Raspberry Pi

Smart Dashcam | Ivy Hub
Hardware side projects: a landmark-aware Raspberry Pi dashcam used on a 2025 Route 66 trip (GPS + OpenStreetMap, offline Piper TTS), and a Flutter Home Assistant hub for the Lenovo Smart Display with a Wyoming voice satellite.
Python, FastAPI, React PWA, Flutter, Kotlin, Home Assistant

brunomoya.com
Portfolio with 66 components, Lighthouse 100 in accessibility, best practices and SEO, and an agent-readable layer (Markdown content negotiation, llms.txt, profile.json).
TypeScript, React, React Three Fiber, Framer Motion, i18next

EDUCATION

BSc in Computer Engineering Universitat Autònoma de Barcelona
2019 - 2023
Barcelona, Spain
Specialisation in Computación (data analysis, statistics, modelling, computer vision). Final degree project awarded Matrícula de Honor.

Génie des technologies de l'information (exchange semester) École de technologie supérieure (ÉTS)
2023
Montreal, Canada
CI/CD, advanced programming and Human-Computer Interaction; coursework in TypeScript.

Batxibac, French-Spanish Double Baccalaureate Institut Gerbert d'Aurillac
2017 - 2019
Barcelona, Spain

CERTIFICATIONS

Hand Tracking Certification ManoMotion 2019
Computer Vision Certification Computer Vision Center 2018

AWARDS & RECOGNITION

Top 50 most promising startups in Catalonia (Glassear) ACCIÓ 2025
Winner, UAB The Hack 2025 (TalentGapAnalyzer) UAB 2025
Matrícula de Honor, final degree project Universitat Autònoma de Barcelona 2023
3rd Prize, ManoMotion game jams ManoMotion 2020, 2021
2nd Prize, Future / Industry 4.0 Maker Faire Galicia 2018
Maker of Merit and special recognition signed by Arduino co-founder David Cuartielles Maker Faire Barcelona 2018

SKILLS

AI / Agentic: Claude Code, OpenAI Codex, GitHub Copilot, LLM APIs, tool & function calling, MCP (FastMCP, remote MCP with OAuth), multi-agent systems, structured outputs, RAG, Langfuse, OpenTelemetry, n8n

Backend: Python, FastAPI, Fastify, Node.js, Rust (Tauri), REST APIs, PostgreSQL, Supabase, Celery, RabbitMQ, Docker, Stripe, PHP/WordPress

Frontend & mobile: TypeScript, React, Next.js, Astro, Vite, Tailwind CSS, Framer Motion, React Three Fiber / three.js, Flutter / Dart, i18next

Engineering: Git, CI/CD, automated testing, API design, system architecture, Linux, monitoring (Sentry), cost optimisation, Proxmox, AWS, Hetzner

XR & hardware: Unity, C#, OpenXR, WebXR, ARFoundation, Android, computer vision, MediaPipe, OpenCV, Arduino, ESP32, Raspberry Pi, 3D printing

Speaking: 20+ talks since 2018, including 4YFN / Mobile World Congress 2025 and 2026, DevFest, Maker Faire and COECYS

Languages: Spanish (native), Catalan (native), English (C2), French (B2)